

XIN YANG

✉ xyang1260@gmail.com  [Xin Yang](#)

EDUCATION

University of Connecticut – <i>Ph.D. in Statistics</i> (Advisor: Dr. Kun Chen)	<i>Aug 2021 - May 2026</i>
Villanova University – <i>M.S. in Applied Statistics</i>	<i>Aug 2019 - May 2021</i>
Nankai University , China – <i>B.S. in Statistics</i>	<i>Aug 2015 - May 2019</i>
Awards: NESS Student Paper Award (2025); ICSA Travel Award (2025)	

PUBLICATIONS

-
1. Yang Xin, Nyiera Hawi N., Sun Yonglei, Zhao Jing, and Chen Kun. Integrative Learning of Intensity Fluctuations of Quantum Dots under Excitation via a Tailored Mixture Hidden Markov Model. *Under revision at Ann. Appl. Stat.*, 2025
 2. Nyiera Hawi N., Yang Xin, Sun Yonglei, Wang Junyu, Liu Yejing, Chen Ou, Chen Kun, and Zhao Jing. An Integrative Statistical Approach for Analyzing Photoluminescence Fluctuations in Single CsPbBr₃ Quantum Dots. *The Journal of Physical Chemistry C*, 2025
 3. Hawi N. Nyiera, Xin Yang, Yejing Liu, Seungjun Cha, Ou Chen, Guoxiang Hu, Kun Chen, and Jing Zhao. Correlating Intensity Fluctuations and Lifetime Distributions Reveals Surface Chemistry Effects on Photoluminescence of Single CsPbBr₃ Quantum Dots. *ACS Applied Materials & Interfaces*, 18(23):32997–33006, 2026

RESEARCH EXPERIENCE

Joint Modeling of Photon Intensity and Lifetime via Coupled Hidden Markov Models <i>UConn, Advisor: Dr. Kun Chen</i>	<i>Sep 2024 - Present</i>
- Developed a coupled HMM to jointly model intensity and lifetime trajectories, capturing their coordinated dynamics - Integrative multi-dot inference that jointly estimates parameters and clusters particles; hypothesis tests validate the findings	

WORK EXPERIENCE

Associate Data Scientist Plymouth Rock Assurance Corporation, Boston, MA (<i>Data Scientist Intern, Jan 2026 - Apr 2026</i>)	<i>May 2026 - Present</i>
- Developed an AI-powered bodily injury claim model to forecast ultimate loss and support accurate reserving - Built an automated ETL pipeline using the OpenAI API to parse unstructured demand packages and claimant questionnaires into structured fields, engineered alongside ten years of historical claim data - Benchmarked XGBoost, Random Forest, and LightGBM; built submodels flagging claims early in the lifecycle - Deployed version-controlled, production-grade Python on AWS SageMaker for automated, repeatable model runs - Partnered with claim adjusters, the claims director, and actuarial to align model outputs with reserving practice	
Summer Data Scientist Intern Plymouth Rock Assurance Corporation, Boston, MA	<i>May 2024 - Aug 2024</i>
- Evaluated external data for catastrophe claim models with XGBoost; proposed a finer flood-risk feature	
CMC Statistician (Co-op) GlaxoSmithKline (GSK), Collegeville, PA	<i>Jan 2021 - May 2021</i>
- CMC statistical support for bioassay transfer and method validation: experiment design, analysis reports, protocol review - Set HCP/DNA residual thresholds via bootstrap and nonparametrics; TOST equivalence testing per ICH Q2(B)	

SKILLS

Statistics & ML:	HMMs, mixture models, EM, survival analysis, bootstrap; XGBoost, LightGBM, LLM-based extraction
Programming & Cloud:	R, Python, SAS, SQL; AWS SageMaker, Snowflake, SQL Server; Git, $\LaTeX$